

Siru Wang

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PROFESSIONAL SUMMARY

Biostatistician and statistical geneticist with expertise in large-scale genomic data analysis for therapeutic target discovery and precision medicine. Experienced in GWAS, multi-ancestry meta-analysis, fine-mapping, and polygenic risk score (PRS) modelling for complex cardiometabolic and renal traits. Proficient in R and Python, with a strong track record of developing scalable, reproducible analytical pipelines in HPC and Linux environments (SLURM). Skilled in integrating diverse genomic datasets, performing variant prioritisation, and translating complex results into actionable insights for multidisciplinary R&D teams.

EDUCATION

2017- 2022 PhD in statistics, school of mathematics and statistics

University of Melbourne, Parkville, VIC3010, Melbourne

Supervisor: A/Prof. Guoqi Qian; co-supervisor: A/Prof. Heejung Shim

2015-2017 M.S in School of Mathematical Sciences

University of Science and Technology of China (USTC), Hefei, Anhui Province

Supervisor: A/Prof. Xiaohong Lan

2011-2015 B.S in Applied mathematics, school of mathematics and statistics

Shandong University, Weihai, Shandong Province

Supervisor: Prof. Yude Bu

PROFESSIONAL EXPERIENCE

➤ **2022/11-Now. MRC biostatistics Unit, University of Cambridge**

Position: research associate (full-time)

Project: Environment-adjusted genetic analysis methods for cardiometabolic traits in African populations

Supervised by Assistant Professor Jennifer Asimit (MRC biostatistics Unit, University of Cambridge)

- Developed an environment-adjusted meta-regression of GWAS, integrating multiple cohort-level summary statistics (>8 cohorts) and varying environmental factors (BMI, sex, urban status), enabling improved detection of genetic associations for lipid traits (LDL-cholesterol, HDL-cholesterol, TG and TC) in underrepresented African populations.
- Developed an environment-adjusted single-trait multi-ancestry fine-mapping method which allows for multiple causal variants to detect the putative causal variants, improving the interpretability of cohort-specific genetic associations.

- Conducted cross-population polygenic score (PGS) transferability analyses, evaluating predictive performance across diverse African cohorts using fine-mapped SNP sets.
- Performed large-scale genomic data harmonization and QC, integrating heterogeneous cohort-level GWAS summary statistics for downstream meta-analysis.
- Produced technical reports and documentation, supporting reproducibility and communication with multidisciplinary teams to interpret results.

Project: Ancestry-specific genetic loci for kidney function identified in the largest African-ancestry meta-analysis

Led by Dr. Oyesola Ojewunmi (Queen Mary University of London, UK)

- Conducted post-GWAS meta-analyses on large-scale African-ancestry datasets, including fine-mapping of genomic risk loci associated with estimated glomerular filtration rate (eGFR) across multiple African populations (East, West and South Africa).
- Prepared comprehensive analysis reports and presented findings to collaborators, ensuring reproducible high-quality outputs suitable for downstream translational research.

➤ **2022/3-2022/11. Agricultural Genomics Institute at Shenzhen, Chinese Academy of Agricultural Sciences (Abbreviation: AGIS)**

Position: postdoctoral fellow (full-time)

Project: Testing for association between gene-by-gene (GxG) and gene-by-environment (GxE) interactions and multiple complex traits.

Supervised by Professor Gengjie Jia (Agricultural Genomics Institute at Shenzhen, Chinese Academy of Agricultural Sciences (Abbreviation: AGIS))

- Developed novel statistical methods within generalized linear models to test association between multiple phenotype traits and GxG/ GxE interactions, addressing limitations in conventional single-trait approaches.
- Applied these proposed methods to European cohort datasets to investigate interaction effects in type 2 diabetes (T2D) and hypertension (HTN), contributing to improved understanding of complex disease.
- Conducted method validation, benchmarking literature-driven optimization, ensuring robustness and alignment with current state-of-the-art techniques.

➤ **2019/4-2019/6. University of Melbourne**

Position: research assistant (part-time)

Project: Empirical study on SLP jurisdiction of the SC: Aggregate analysis

Supervised by Professor Tarunabh Khaitan (University of Melbourne)

- Brand insights analytics and data visualization: utilize jurisdiction data to generate hypothesis, test assumptions and explore jurisdiction insights.

➤ **2014/3-2014/5. Shandong University, Weihai**

Position: research assistant (part-time)

Project: Astrophysics Parameter Measurement

Supervised by Professor Bu Yude (Shandong University, Weihai)

- Coding in matlab and writing the proposal of the project "Astrophysics parameter measurement".

PROGRAMMING & SOFTWARE

R packages development (statistical genetics & genomic analysis)

- ***varSelRFMis***

The R package *varSelRFMis* is a coherent algorithm of categorical phenotype-genotype association analysis, in the presence of missing values with non-ignorable missingness mechanism in genomic data.

- ***env.MRmega***

The R package *env.MRmega* is designed for identifying genetic variants that are associated with each trait, across cohorts, while adjusting for differing environmental exposures between cohorts.

- ***envMRMEGAfm***

The R package *envMRMEGAfm* (env-MR-MEGA fine-mapping) is designed for simultaneously fine-mapping genetic associations across multiple cohorts, allowing for multiple causal variants.

- ***GiMat***

The R package *GiMat* (Gene Interaction and Multiple-phenotype Association Test) is designed for joint analysis of multiple phenotypes with gene-by-gene (GxG) interactions, incorporating correlation structures between multiple traits to improve detection power.

- ***mrmegaPRSpip (in preparation for release)***

Designed an end-to-end PRS construction pipeline integrating:

- GWAS meta-analysis for locus discovery
- Fine-mapping for causal variant prioritization
- Polygenic score (PGS) calculation using PLINK and PRSice software.

Focused on multi-ancestry PGS development and transferability.

Programming & Computational skills

- **Languages:** R, Python, C/C++ (via Rcpp)
- **Genomics tools:** PLINK, fine-mapping methods (SuSiex, MESuSiE, MultiSuSiE), meta-analysis method (METAL, MR-MEGA)
- **Infrastructure:** High-performance computing (HPC) environments.
- **Reproducibility:** R package development, documented pipelines.

RESEARCH EXPERTISE

- Genome-wide association studies (GWAS)
- Fixed-effect, random-effect, and multi-ancestry meta-analysis
- Multi-ancestry and multi-trait fine-mapping for causal variants identification
- Polygenic risk scores (PGS) development and evaluation for complex traits and diseases
- Multi-omics data integration
- Gene-environment (GxE) and Gene-gene (GxG) interaction association studies
- Rare variant and common variant association analysis

PUBLICATION AND PREPRINTS

- Wang, S., Qian, G., & Hopper, J. (2023). Integrated Logistic Ridge Regression and Random Forest for Phenotype-Genotype Association Analysis in Categorical Genomic Data Containing Non-ignorable Missing Values. *Applied Mathematical Modelling*
- Wang, S., Ojewunmi, O.O., Kamiza, A. et al. (2024). Accounting for heterogeneity due to environmental sources in meta-analysis of genome-wide association studies. *Commun Biol* 7, 1512
- Wang, S., Ojewunmi, O., Pirie, F., Motala, A., Ramsay, M., Morris, A., Fatumo, S., Chikowore, T. and Asimit, J.L., (2025). Multi-ancestry fine-mapping accounting for ancestral and environmental heterogeneity improves resolution and interpretation. *bioRxiv*, pp.2025-11.
- Wang, S., Zhu, X., Li, Y. and Jia, G., (2025). A statistical framework to identify gene-gene interactions underlying multiple dichotomous phenotypes from genotype data. *medRxiv*, pp.2025-11

CONFERENCES AND SEMINARS

- **Online attendance**, European Mathematical Genetics Meeting (EMGM), April 20-21st, 2023
- **Oral presentation**, BSU Together seminar, MRC biostatistics unit, University of Cambridge, May 2023.
Title: Environment-adjusted genetic association methods for cardiometabolic traits in African populations
- **Oral presentation**, Causal mechanism theme meeting, MRC biostatistics unit, May 2024.
Title: Accounting for heterogeneity due to environmental sources in meta-analysis of genome-wide association studies.
- **Oral presentation** (selected as the Early Career Awards candidate and awarded ESHG2024 conference fellowship), The European Human Genetic Conference (ESHG), Berlin Germany, June 1-4th, 2024.
Title: Adjusting for environmental factors in multi-ancestry meta-regression improves power to detect genetic associations
- **Poster presentation**, The 15th African Society of Human Genetics Conference and the 1st Ugandan Society of Human Genetics and Bioinformatics (AfSHG), Entebbe Uganda, Feb 3-7th, 2025.
Title: Accounting for heterogeneity due to ancestry and environment improves the resolution of multi-ancestry fine-mapping
- **Oral presentation**, European Mathematical Genetics Meeting (EMGM), Brest France, April 8-9th, 2025
Title: Multi-ancestry fine-mapping accounting for ancestral and environmental heterogeneity improves resolution
- **Oral presentation** (selected as the Early Career Awards candidate), The European Human Genetic Conference (ESHG), Milan Italy, May 24-27th, 2025.
Title: Multi-ancestry fine-mapping accounting for ancestral and environmental heterogeneity improves resolution

- **Oral presentation** (selected as a candidate for Neel Award Finalist), International Genetic Epidemiology Society Annual meeting (IGES), Cologne, Germany, Sep 1-2nd, 2025.
Title: Heterogeneity Due to Ancestry and Environment Improves the resolution of multi-ancestry fine-mapping
- **Poster presentation**, The American Society of Human Genetics Annual Meeting (ASHG), Boston, US, Oct 14-18th, 2025.
Title: Multi-ancestry fine-mapping accounting for ancestral and environmental heterogeneity improves resolution and interpretation
- **Oral presentation**, BSU Together seminar, MRC biostatistics unit, University of Cambridge, Dec 17th, 2025.
Title: Accounting for Heterogeneity Due to Ancestry and Environment Improves the Resolution of Multi-ancestry Fine-mapping

TEACHING EXPERIENCE

Teaching assistant

- **2016** Mathematical statistics, University of Science and Technology of China (USTC)
- **2018** MAST90084 Statistical modelling, University of Melbourne (UoM)
- **2018-2022** MAST20005 Statistics, University of Melbourne (UoM)
- **2020** MAST20006 Probabilities for statistics, University of Melbourne (UoM)